

Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—AI data centers can provide demand response (DR), yet a meter decrease does not show that the workload was committed before the event or that the change had positive network value. This paper develops a declaration-indexed verifier for spatially coupled data centers. Submit-time scheduler records define release, runtime, GPU request, class, and region; a cumulative workload linear program, a convex risk contract, and a contiguous runtime witness share this ledger and its digest. An active-window energy map prevents carry-in declarations from being omitted when the aggregate contract is scaled to locked days. The risk fit enforces total and daily conditional value at risk (CVaR) limits together with a predeclared validation-error non-inferiority certificate. For the 54-day locked block, the causal Pareto contract reaches nRMSE 0.691, 1.895 MWh/day false credit, 17.024 MWh/day under-credit, and $F_1 = 0.699$, weakly dominating the declaration-causal workload-feasible comparator set (with strict gains over the single feasible projection). The executable bridge enumerates 7,306,622 declaration-feasible starts, reproduces the optimum within 2.3×10^{-13} USD, and obtains total-profile nRMSE 0.036 while reporting the flexible relative-L2 residual 1.410. A signed RTS-24 N-1 replay is kept separate from the independent observed-meter scoring panel; it is not labeled as a payable closed-meter settlement. The evidence therefore distinguishes declaration feasibility, simulated mechanism isolation, and observational scoring, and does not identify a causal utility treatment effect.

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known

bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide this physical state: temporal deferral and geographical balancing reduce energy cost and peak demand [5]. Data-driven flexibility and security-constrained coordination optimize spatial-temporal dispatch [5], [6]. Balancing-market and load-shifting studies provide complementary dispatch contexts. Virtual-link clearing and space-time remuneration model geographical and temporal transfers [7], [8]. Receding-horizon frequency regulation retains multi-site state and future arrivals [9]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains open. Coupled-regulation models likewise provide no provenance-bound certificate [10]. Whole-facility measurements also require a bottom-up GPU-to-facility conversion; calibration, benchmark scaling, and network placement are therefore kept as separate evidence layers.

The second gap is electrical. Geographic workload control is often valued with energy prices or gross megawatts, although nodal marginal value changes when congestion or a contingency binds. Direct-current (DC) power flow and linear distribution factors provide transparent sensitivities [11], [12]; N-1 dispatch imposes credible post-contingency limits through line-outage distribution factors (LODFs) [13]. No data-center DR verifier jointly certifies workload feasibility, signed spatial response, and security-aware network value. Alternating-current (AC) feasibility is a separate nonlinear check. Public traces support observational replay but not a causal utility event, so measured execution, counterfactual construction, and network valuation must remain separate.

This paper makes three power-system contributions. 1) It defines a typed submit-execute-settle state in which a runtime-complete job witness is formed from release, declared runtime, GPU request, and regional assignment. An active-window energy map gives the exact declaration energy intersecting a locked operating window, including carry-in jobs. The same indexed digest is rechecked at regional aggregation and reused by the RTS-24 N-1 value replay. 2) It formulates a workload-state verifier with a causal information boundary, a validation-

selected high-5-of-10 Pareto projection, and a convex total-plus-daily-CVaR contract. A triangle-inequality certificate constrains the validation RMS error to a predeclared tolerance; in the locked fit it selects the exact causal Pareto vertex while the epigraph controls absolute false-credit exposure. The aggregate contract is then mapped to a fixed linear dual-price surrogate and an exact finite contiguous-block policy; primal target residuals remain explicit. 3) It gives the workload response a signed nodal value under complete finite N-1 contingencies and separates declaration feasibility, network-value replay, and meter-gated payment activation. This separation prevents a declaration trajectory from being interpreted as a closed-meter payment. The resulting coupling invariant carries one declared workload state into physical verification and security-aware valuation; its LP, dispatch, and allocation primitives retain established power-system semantics [14]. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature-stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler provenance: a digest detects record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The submit-time declaration ledger supplies arrival energy a_{skt} for source s , class k , and release interval t . The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$ and deadline D_k , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion is $y_{skt} - y_{sk,t-1} = \sum_d x_{skd\tau}$ with $y_{sk,-1} = 0$, and terminal service equals the declared arrivals due in the window; the release and deadline inequalities are (2) and (3), and the site-capacity constraint is $\sum_{s,k} x_{skd\tau} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skd\tau}. \quad (4)$$

For rolling window end T^+ and final event-day release T^0 , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk, \max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival while retaining later real arrivals. The window starts $D_{\max} + 96$ intervals before and ends $D_{\max}$ intervals after the event; omitted earlier work has expired. Equations (2)–(4) map an aggregate preemptive envelope to the facility meter. This is valid for checkpointable batch work; Experiment 14 separately audits contiguous execution intervals, runtime, GPU count, and native power. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, r_j denote its submitted release, and let d_j be the service-window endpoint. Slurm “timelimit” is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_{\infty} = 128$ slots before adding B ; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is fixed before the split. The training-calibrated entitlement $\hat{E}_j$ is retained only for the aggregate Exp19 audit; the executable witness uses the declared runtime. Let $\bar{p}_{\text{GPU}} = 0.001$ MW/GPU be the committed nameplate, s_j the submit-time scenario region, and g_j^{req} the requested GPU count. Define the runtime-complete entitlement $E_j^{\text{run}} = R_j g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$. The indexed service variable $u_{j\tau}$ is the energy assigned to job j in interval τ , and the executable witness requires one contiguous fixed-rate block of exactly R_j slots. Let w_{batch} be the predeclared waiting-cost coefficient per MWh-slot, and r^{DR} the event service tariff. Define $c_j = g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$ and $\Theta_j = \{r_j, \dots, d_j - R_j\}$. For each admissible start θ , a binary $z_{j\theta}$ selects the block and $u_{j\tau}^{\theta} = c_j \mathbf{1}\{\theta \leq \tau < \theta + R_j\}$. The exact start-time mixed-integer linear program (MILP) is

$$\begin{aligned} \sum_{\theta \in \Theta_j} z_{j\theta} &= 1, \quad z_{j\theta} \in \{0, 1\}, \\ u_{j\tau} &= \sum_{\theta \in \Theta_j} u_{j\tau}^{\theta} z_{j\theta}, \quad \sum_{j: s_j = d} u_{j\tau} \leq \bar{P}_d \Delta t, \\ \min_z \sum_{j, \theta \in \Theta_j} z_{j\theta} \sum_{\tau} [w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\}] u_{j\tau}^{\theta}. \end{aligned} \quad (6)$$

Every selected block enforces runtime completion, contiguity, and the GPU nameplate; its positive regional slack makes the separable enumeration exact for the displayed MILP. For delay $\delta = \theta - r_j$, the waiting part of the objective is $w_{\text{batch}} c_j \{R_j \delta + R_j(R_j - 1)/2\}$, so a long job is charged for every delayed service interval rather than only for its start time. Experiment 27 enumerates every admissible start for all 75,326 records and stores the earliest-start baseline and tariff response; Experiment 25 adds a binding regional row. The measured $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is opened only for independent post-event scoring after the declaration witness is fixed: 71,128 execution matches never constrain z , u , or d_j . Thus $\hat{E}_j$ is calibration-only, E_j^{run} is the nameplate-capacity commitment, and the calibrated central energy is carried on the same blocks without being presented as measured consumption. The risk target is linked through a fixed regional slot price. Because the

statistical and nameplate ledgers have different scales, define the locked window set $\mathcal{W}_{\text{lock}} = \bigcup_{h \in \mathcal{H}_{\text{lock}}} \{96h, \dots, 96h + 95\}$. The declaration energy active in this set is

$$E_{\text{act}}(\mathcal{W}_{\text{lock}}) = \sum_{j \in \mathcal{J}_{\text{sub}}} E_j^{\text{dec}} \frac{|[r_j, r_j + R_j] \cap \mathcal{W}_{\text{lock}}|}{R_j}, \quad (7)$$

where E_j^{dec} is either the nameplate upper entitlement or the calibrated central entitlement. This overlap integral includes carry-in jobs and uses only submit-time fields. Let $\tilde{p}_{dht}^{\text{risk}} = P_d^{\text{fix}} + \gamma_+ [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$, with $\gamma_+ = E_{\text{act}}(\mathcal{W}_{\text{lock}}) / \sum_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ \Delta t$. For locked day h , region d , and interval t , define the fixed linear dual-price surrogate $\pi_{dht} = r^{\text{DR}} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ / \tilde{\Pi}_{\text{max}}$, where $\tilde{\Pi}_{\text{max}} = \max_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$. Re-enumerating every declaration-feasible block with the waiting, event-tariff, and π_{dht} terms produces the exact minimizer of this finite linear-price objective. The optimality statement concerns the declaration-feasible policy; equality to the aggregate target is a separate primal residual. The bridge therefore records, for every locked day,

$$e_{dht}^{\text{br}} = p_{dht}^{\text{exec}} - \tilde{p}_{dht}^{\text{risk}}, \quad (8)$$

$$\text{nRMSE}_{\text{flex}} = \frac{\|e^{\text{br}}\|_2}{\max\{\|\tilde{p}^{\text{risk}} - P^{\text{fix}}\|_2, \epsilon\}}.$$

along with the total-load normalization and the flexible relative-L2 denominator. Experiment 29 independently re-enumerates 7,306,622 candidate starts, verifies the stored minimizer, and reports these residuals. The bridge stores raw/scaled profiles, the common digest, daily residuals, and the central-energy check; it is a deterministic realization of the fitted target under the declared surrogate, not a second aggregate optimization. For the job-to-network certificate, the indexed witness is mapped without a second optimization. To avoid overloading the deadline endpoint d_j , let $\mathcal{J}_{sk\delta}$ be the ledger partition whose source, class, and declared destination labels are s, k, δ ; Experiment 27 uses the native assignment $\delta = s_j$ and therefore has no outcome-dependent migration:

$$x_{sk\delta\tau}^{\text{job}} = \sum_{j \in \mathcal{J}_{sk\delta}: r_j \leq \tau < d_j} u_{j\tau}, \quad (9)$$

$$p_{\delta\tau}^{\text{job}} = P_{\delta}^{\text{fix}} + \Delta t^{-1} X_{\delta\tau}^{\text{job}}, \quad X_{\delta\tau}^{\text{job}} = \sum_{s,k} x_{sk\delta\tau}^{\text{job}}. \quad (10)$$

Aggregate migration remains a matched-workload control; (10) is the indexed replay profile. Experiment 27 reconstructs X^{job} from every stored $u_{j\tau}$ and verifies equality before network valuation; no separately optimized aggregate receives a network value. Workload deferral and geographic assignment follow [5], [6], while rolling-state control follows [9]; waiting and migration costs are $w_k(\tau - t)x_{sk\delta\tau}$ and $m_{sd}x_{sk\delta\tau}$, with service-time substitution differing only by a fixed-arrival constant.

The submit-time and post-event ledgers have separate canonical digests. Each submitted record is represented by the scheduler-only tuple; its Secure Hash Algorithm 256 (SHA-256) digest is

$$\begin{aligned} \ell_j^{\text{sub}} &= (\text{id}_j, t_j^{\text{submit}}, \text{timelimit}_j, g_j^{\text{req}}, \kappa_j, \sigma_j), \\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell_j^{\text{sub}} : j \in \mathcal{J}_{\text{sub}}\}, \\ h_{\text{sub}} &= \text{SHA-256}(\text{serialize}(\mathcal{L}_{\text{sub}})). \end{aligned} \quad (11)$$

where g_j^{req} , κ_j , and σ_j are the requested-GPU, class, and scheduler-state fields. The resulting canonical commitment is

(11). Start/end times, DCGM energy, and telemetry counts enter a separate execution ledger whose join and energy equality are checked only after commitment. Canonical ordering makes h_{sub} independent of execution telemetry [15]; Experiment 16 reconciles measured power with the 118-MW nameplate fixed before the split.

a) *Information and threat model.*: At decision time, $\mathcal{I}^{\text{dec}}$ contains the submission ledger, declared topology, pre-event prices, and configuration; the scoring set adds the closed meter and execution ledger only after the contract is frozen. The threat model covers post-commitment edits, replayed records, and attempts to introduce future arrivals into a committed response. The canonical digest rejects an altered scheduler tuple, the information-boundary certificate rejects future or locked-test fields, and the settlement rule caps payment with the closed meter. Scheduler and DCGM services remain trusted sources: a digest detects a changed field, while upstream attestation is required to establish that a submitted deadline, region, class, or energy field was semantically truthful. Complete-ledger profiles are therefore ex-post information comparators; the slot-62 committed ledger is the deployable contract protocol.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p denotes regional power in MW and time sums use Δt (MWh). The declaration-causal estimate is $\tilde{p}$; p^0 is the declaration-only strategic reference used for mechanism isolation, $p^{1,\text{sim}}$ is the independently solved event-response schedule, and p^{obs} is the Massachusetts Institute of Technology (MIT) Data Center GPU Manager (DCGM) execution trace. The observed trace is opened only after a profile is frozen; no utility event label exists, so it is never a causal input. The mechanism-isolation response $r = p^0 - p^{1,\text{sim}}$ is signed and retains rebound. For $z \in \{\text{obs}, \text{sim}\}$, submitted and true credit are $\hat{r}^{(z)} = [\tilde{p} - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{r}_{dt}^{(z)} - r_{dt}^{(z)}]_+, \quad z \in \{\text{obs}, \text{sim}\}. \quad (12)$$

The simulated source defines the validation epigraph; the observed source is recomputed only in locked replay. False credit is computed from (12) in absolute MW-slots, avoiding a ratio with a zero-credit denominator. Before the event the operator freezes a contract baseline p^{con} . After meter closure, define the submitted gross reduction, the contract-cap reduction, and the payable reduction by

$$\begin{aligned} q_{dt}^{\text{gross}} &= [\tilde{p}_{dt} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{cap}} &= [p_{dt}^{\text{con}} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{pay}} &= \min\{q_{dt}^{\text{gross}}, q_{dt}^{\text{cap}}\}. \end{aligned} \quad (13)$$

The capacity payment is $Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{pay}}$. Equation (13) uses only the submitted baseline, frozen contract, and closed meter. Here $p^{\text{con}} = p^{\text{safe}}$ for the payment-band panel and is the no-tariff committed-ledger profile for the decision-time panel; p^0 remains an offline diagnostic. On locked event slots, $\text{nRMSE} = \sqrt{N_E^{-1} \sum_n (\tilde{p}_n - p_n^0)^2} / \max\{N_E^{-1} \sum_n p_n^0, \epsilon\}$. With $M = \sum_n \min\{\hat{r}_n, r_n\}$, precision, recall, and F_1 are $M / \max\{\sum_n \hat{r}_n, \epsilon\}$, $M / \max\{\sum_n r_n, \epsilon\}$, and $2PR / \max\{P + R, \epsilon\}$ over the N_E region-slot samples.

The reported error metrics use mean absolute error (MAE) and root-mean-square error (RMSE); normalized RMSE (nRMSE) is the scale-free form above. The risk simplex is the sole convex quadratic program (QP); all released workload counterfactuals and payment caps use the linear programs stated above.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q , and DR price π^{DR} . If reference day j adds event-window service δR_j , the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (14)$$

Let c'_j be the right derivative of minimum operating cost with respect to the event-window service constraint on day j . Linear-program sensitivity [16] gives

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (15)$$

At a degenerate optimum the current dual can be zero while the right marginal cost is positive; Experiment 1 therefore checks c'_j with independent 5×10^{-4} and 10^{-3} MWh finite differences (agreement tolerance 10^{-6} USD/MWh). Equations (14)–(15) specialize the historical-baseline premise in [4].

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\text{native}} + Mp_t^{\text{dc}}$ the nodal demand vector. For generator-to-bus map C_g , generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \sum_i C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad g \leq \bar{g}, \\ -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \quad (16)$$

The incidence mapping aggregates demand at its connection bus; (16) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (17)$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. PROPOSED METHODOLOGY

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations define the feasible state before a payment target is formed (Fig. 1). Meter history and submitted declarations enter the estimator in parallel; execution remains scoring-only. Lagged meter/calendar baselines and declaration-view ledger comparators include ridge, gradient boosting, and extremely randomized trees [17], [18]. Synthetic control and nonnegative donor weighting are evaluated as a separate history-based class [19], [20]. The quantile output receives the same exact projection, with its penalty selected by contiguous folds.

B. Sparse cumulative workload-state verifier

For statistical estimate $\tilde{p}$ and each predeclared penalty ρ_ℓ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (18)$$

subject to (1)–(4). The absolute-value epigraph is standard [16]; its use for workload-ledger verification is specific here. C_w is workload cost, C_i is the generator-segment cost in (16), and $V(p)$ is dispatch value, so (18) changes only the objective. Six globally solved metadata schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh causal Pareto projection is selected on validation from the high-5-of-10 declaration target by a lexicographic rule over nRMSE, false credit, under-credit, and $1 - F_1$. The risk simplex contains these seven causal profiles. A complete-ledger feasible-quantile projection is retained as an external post-commitment comparator and is used only in the separate payment hull. For a hull profile, $\bar{p} = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation error plus total and daily-tail false-credit exposure, measured by conditional value at risk (CVaR) [21]. Let ℓ^* minimize contiguous-fold nRMSE and let B_h be the absolute daily false-credit exposure of that single feasible projection on validation day h . For a candidate simplex profile define $F_h(\alpha) = \sum_{n \in h} f_n$, where f_n is the positive-part false credit at sample n . The risk fit uses absolute MW-slot exposure throughout; there is no ratio with a potentially zero true-credit denominator. With $m = \lceil (1 - \gamma)H \rceil$ and reserve $\beta \in (0, 1]$, solve

$$\begin{aligned} \min_{\alpha, f, \nu, u} \quad & \frac{\|\bar{p} - p^{0, \text{decl}}\|_2^2}{Ns_p^2} + \lambda \|\alpha\|_2^2 + \omega_T \frac{\sum_{h=1}^H F_h}{S_T} \\ & + \omega_C \frac{\nu + m^{-1} \sum_{h=1}^H u_h}{S_C}, \end{aligned} \quad (19)$$

Equation (19) fixes both preferences inside the fit. Here $s_p^2 = \max\{N^{-1}\|p^{0, \text{decl}}\|_2^2, \epsilon\}$, $S_T = \max\{\sum_h B_h, \epsilon\}$, and $S_C = \max\{m^{-1} \sum_{h \in \text{tail}_\gamma} B_h, \epsilon\}$. Single-axis ablations remove one term and its constraint; the locked joint fit uses $\omega_T = \omega_C = 0.50$, $\gamma = 0.75$, and $\lambda = 10^{-8}$. The budgets are

$$\sum_{h=1}^H F_h(\alpha) \leq \beta \sum_{h=1}^H B_h, \quad (20)$$

$$\text{CVaR}_{0.75}(F_h(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_h). \quad (21)$$

The positive part is represented without differentiating through a sort. For each event sample n and validation day h , introduce $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_h \geq 0$:

$$f_n \geq p_n^\alpha - p_n^{(\text{sim})} - r_n^{(\text{sim})}, \quad (22)$$

$$F_h(\alpha) = \sum_{n \in h} f_n, \quad (23)$$

$$F_h(\alpha) - \nu - u_h \leq 0, \quad (24)$$

$$\nu + \frac{1}{m} \sum_h u_h \leq \beta \text{CVaR}_{0.75}(B_h). \quad (25)$$

The linear CVaR epigraph in (25) uses the same m -day denominator as the empirical statistic and avoids sorting. Here $r_n^{(\text{sim})} = [p_n^0 - p_n^{(\text{sim})}]_+$; the implementation uses the equivalent threshold $p_n^{1, \text{sim}} + r_n^{(\text{sim})}$ and stores the identity audit. To preserve the point-estimate metric, let $q_\ell = N^{-1/2}\|p^{(\ell)} - p^{0, \text{decl}}\|_2$

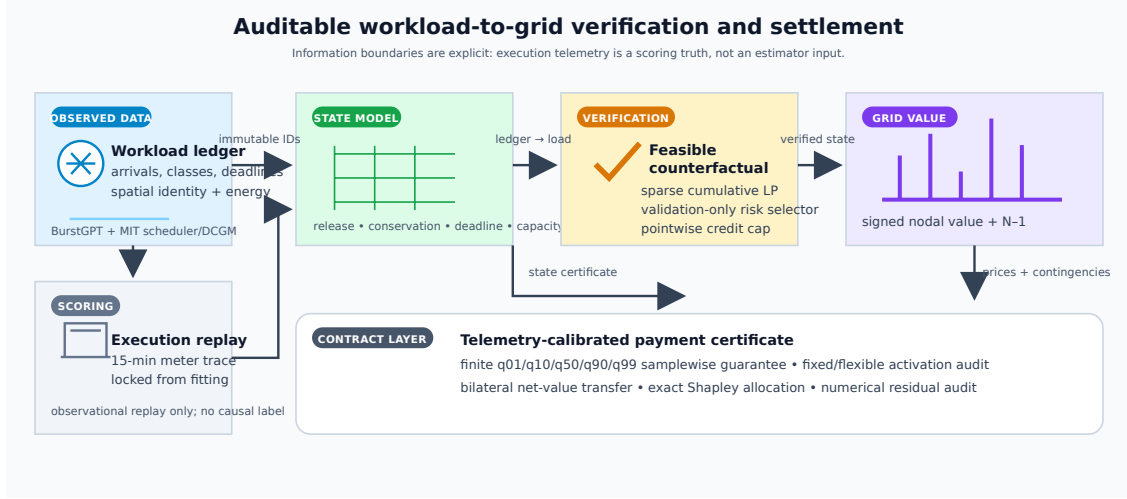


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N-1 value and exact participant shares.

and q_* denote the RMS error of the validation-selected causal Pareto reference. The simplex additionally satisfies the linear sufficient condition

$$\sum_{\ell=1}^L \alpha_{\ell} q_{\ell} \leq (1 + \varepsilon_{\text{acc}}) q_*, \quad \varepsilon_{\text{acc}} = 10^{-6}. \quad (26)$$

By the triangle inequality, (26) bounds the actual simplex RMS error by the same right-hand side; it therefore prevents a risk improvement from being purchased with an unreported forecast degradation. Because the validation-selected causal Pareto profile is the minimum-RMS member of this declared simplex, the zero-width certificate has an exact closed-form optimizer: the one-hot causal Pareto vertex. This is the solution of the same convex quadratic program, not a post-solution rule. Independent Karush–Kuhn–Tucker stationarity and epigraph residuals are recomputed after the solve; simplex, accuracy, and primal residuals are checked as well. Four contiguous folds select β only when each training contract is feasible. A separately reported payment-contract projection targets $\bar{p}$ while imposing

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell*)}. \quad (27)$$

The pointwise cap in (27) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

a) Two-sided credit band. To exclude a zero-credit envelope, tie a lower band to the fitted target. Define $p_{dt}^{\text{floor}} = \min\{p_{dt}^{(\ell*)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the pre-projection validation fit. The event-slot constraints are

$$p_{dt}^{\text{floor}} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (28)$$

The release sets $\varepsilon = 0$; the lower side is therefore the exact pointwise minimum of the two independently formed profiles, and the selected single projection proves nonemptiness. Equations (20)–(21) control aggregate and daily-tail exposure, while the lower side in (28) regularizes under-credit and gives no unseen-trace guarantee. Realized overpayment is handled by (13).

b) Decision-time information boundary. A gate removes all post-gate arrivals before refitting the matched-information target. The locked protocol uses slot 62, 30 minutes before the first event slot, and a no-tariff LP freezes the contract baseline with a predeclared 3-MWh minimum of participating event-window service. A second LP applies the selected positive DR price to the same masked ledger while carrying unexpired state. Reserved future jobs cannot be credited, and the tariff plan is scored against the closed meter through (13). The independent execution trace is read only in the scoring pass.

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d ,

$$\hat{a}_{sk\tau}^{\text{res}}(d; \eta) = Q_{\eta}(\{a_{sk\tau}(h) : h < d\}), \quad \tau \geq t_{\text{gate}}, \quad (29)$$

Equation (29) uses earlier days only. The predeclared candidate set is $\{0.55, 0.60, 0.65, 0.70\}$ with a 1.0-MWh/day validation false-credit allowance; it selects $\eta = 0.55$ before locked scoring. The reserve is a planning envelope, and no reserved job enters payment before commitment.

The cumulative state has $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$ nonzeros, versus quadratic explicit prefixes. Equation (5) retains every observed future arrival for the 512-slot deadline. Two lexicographic LPs (minimum L1, then operating cost) embed the day trajectory and are solved globally with HiGHS [22]. The resulting state-to-credit path is shown in Figs. 1 and 2; execution and interval audits are separate (Experiments 13–15).

C. Signed nodal and N-1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^{\top}(p^0 - p^1)$, while (17) is the realized signed system-value benchmark. Convexity [16] gives

$$0 \leq (\lambda^0)^{\top}(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^{\top}(p^1 - p^0). \quad (30)$$

Thus the linear rule upper-bounds exact value in the polyhedral market model; equality requires a common supporting subgradient.

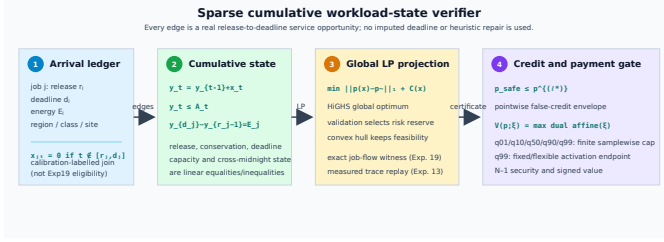


Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from anticipatory shift through recovery. Its space-time remuneration is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_d \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \quad (31)$$

This is the nodal virtual-link structure of [7], [8]; summing site payments reproduces (31).

Space-time energy remuneration is one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^0 - p_{dt}^1), \quad (32)$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \quad (33)$$

where p^0 is the minimum-cost no-event schedule on the same window. Equation (32) defines the bilateral capacity product and need not equal trace-anchored p^0 ; Π^{ST} debits spatial transfer and recovery. This signed service model follows demand-response mechanism design [23], [24].

Let x^1 and x^0 induce p^1 and p^0 . Define the participant opportunity cost $C_{\text{opp}} = \mathcal{C}_w(x^1) - \mathcal{C}_w(x^0)$ and surplus $S = G - C_{\text{opp}}$. With zero disagreement utilities, the symmetric Nash contract [25] is

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\text{B}} = z \left(C_{\text{opp}} + \frac{S}{2} \right), \quad u_{\text{dc}} = u_{\text{op}} = z \frac{S}{2}. \quad (34)$$

Equation (33) combines the capacity payment and signed space-time value. Equation (34) uses nonactivation as the outside option, keeps both utilities nonnegative, and is budget balanced. It is separate from ISO nodal remuneration and the exact dispatch-value benchmark. For the executable full-day ledger, each interval first retains the signed contract value $s_{dt} = r^{\text{DR}} q_{dt}^{\text{pay}} + \Delta V_{dt}$, where power is in MW, Δt is in hours, and both terms are in USD. The reported economic ledger is $S^{\text{signed}} = \sum_{d,t} s_{dt}$; a cash presentation $S^{\text{cash}} = \sum_{d,t} \max\{s_{dt}, 0\}$ is stored separately. The cash floor therefore leaves negative network-value debits visible in the signed cycle total. The complete-cycle ledger $\mathcal{W}$ includes prevent increases and rebound, so workload conservation gives $\sum_{d,t \in \mathcal{W}} (p_{dt}^0 - p_{dt}^1) \Delta t = 0$ within solver tolerance.

Before payment comparison, the submitted, frozen-contract, and closed-meter reductions are intersected as follows:

$$q_{dt}^{\text{pay}} = \min\{q_{dt}^{\text{obs}}, q_{dt}^{\text{cap}}\}, \quad G = r^{\text{DR}} \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{pay}} + \Pi^{\text{ST}}. \quad (35)$$

Equation (35) is evaluated after the submitted profile, contract, and closed meter are fixed; the second line uses q^{pay} in the corresponding ledger. Experiment 12 stores the observed meter chain and its budget residual (below 10^{-8} USD). Experiment 27 instead stores the declaration replay $q_{dt}^{\text{replay}} = \min\{q_{dt}^{\text{dec}}, q_{dt}^{\text{cap}}\}$ to certify feasible network value. Exp30 records this role as declaration-only, and the replay's signed/cash totals remain outside payable settlement until q^{obs} is available.

Security-aware settlement augments (16) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ , then

$$-\bar{f}_{\ell} \leq (H_{\ell} + L_{\ell k} H_k)(C_{gg} - p) \leq \bar{f}_{\ell}, \quad \ell \neq k. \quad (36)$$

Equation (36) is the standard preventive N-1 LODF formulation [13]; every credible row uses native ratings. Islanding rows are excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the DC/N-1 planning scope [26].

D. Scenario-robust N-1 payment certificate

The pointwise envelope controls energy but not network payment because spatial loads can activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the feasible-profile hull and $p^{\text{cap}} \in \mathcal{P}$ the validation-selected projection. Its penalty is separate from the payment-target penalty; the feasible-quantile projection is an external comparator. Let Ξ_5 be the 1st, 10th, 50th, 90th, and 99th calibration-validation GPU-energy ratios from independent telemetry [2]. For facility profile p_{dt}^{fac} , fixed demand P_d^{fix} , scale s_{dc} , and factor ξ ,

$$p_{dt}^{\text{dc}, \xi} = s_{\text{dc}} (P_d^{\text{fix}} + \xi [p_{dt}^{\text{fac}} - P_d^{\text{fix}}]), \quad p_{dt}^{\text{fac}} \geq P_d^{\text{fix}}. \quad (37)$$

The scale uses the calibration-validation q_{99} flexible peak, keeping the fixed component invariant to ξ ; raw endpoints remain activation scenarios. The injection is $M p_t^{\text{dc}, \xi}$, with superscripts α and “cap” applying (37) to p^{α} and p^{cap} . For each settlement day, the payment-certified verifier defines $B_{\xi}^{\text{cap}} = \Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \text{cap}})$ and solves

$$\text{lexmin}_{\alpha, e, \eta, \{g_t^{\xi}\}} \left(\sum_{d, t \in \mathcal{E}} e_{dt}, -\eta \right) \quad (38)$$

$$\begin{aligned} \text{s.t. } & p^{\alpha} = \sum_{\ell=1}^L \alpha_{\ell} p^{(\ell)}, \\ & \alpha \geq 0, \quad \mathbf{1}^{\top} \alpha = 1, \quad 0 \leq \eta \leq 1, \\ & e_{dt} \geq \pm (p_{dt}^{\alpha} - p_{dt}^{\text{safe}}), \\ & g_t^{\xi} \text{ satisfies (16) and (36),} \\ & \text{with } p_t = p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \alpha}, \\ & \quad t \in \mathcal{E}, \quad \xi \in \Xi_5, \\ & \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^{\xi}) \\ & \quad \leq (1 - \eta) B_{\xi}^{\text{cap}}, \quad \xi \in \Xi_5. \end{aligned}$$

The piecewise-linear epigraph is solved by two global LPs: trajectory error is minimized first, then the common fractional margin η . Feasibility is the N-1 upper bound and p^{cap} proves nonemptiness. Hence, for any $\xi \in \Xi_5$ and realized load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned} & \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\alpha}) - V_{N-1}(p_t^{1,\xi})] \\ & \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\text{cap}}) - V_{N-1}(p_t^{1,\xi})]. \end{aligned} \quad (39)$$

The realized cost cancels, so this is a samplewise certificate without execution truth. Every finite non-islanding contingency is imposed using standard SCED/LP primitives [12], [16]. The hull contains six first-stage projections and the matched feasible-quantile comparator. risk-constrained verifier is the validation-fitted simplex point under the two risk budgets; the separately stored payment contract targets it under the pointwise cap and is excluded from risk fitting.

The finite cap also has a set-valued certificate. Let $p^{\text{cap}} = p^{(\ell^*)}$, p^{pay} be the separately selected payment target, and $p^{\text{cert}} = p^\alpha$. For $\Delta V_\xi(p) = V_{N-1}(p, \xi) - V_{N-1}(p^{1,\xi})$ and $p_\theta = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$, the interval is

$$\mathcal{I}_\xi = \left[\min_{0 \leq \theta \leq 1} \Delta V_\xi(p_\theta), \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\} \right]. \quad (40)$$

The lower endpoint in (40) is one joint N-1 SCED LP with a shared segment parameter; convexity gives the endpoint maximum and the cap belongs to $\mathcal{I}_\xi$. Validation-frozen profiles and the realized counterfactual determine both endpoints; the interval width is uncertainty cost, while (38) remains the finite-scenario certificate.

E. Participant allocation

For participants N , let $v(S)$ be signed dispatch value when coalition S moves from baseline to actual demand. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (41)$$

This Shapley specialization in (41) uses signed grid value [27]; coalition-size grouping gives $\sum_i \phi_i = v(N)$ when $v(\emptyset) = 0$. All 16 four-site and 256 eight-portfolio coalitions are enumerated exactly. For larger contracts, exchangeable site slices retain binomial multiplicities and reduce evaluation to $\prod_g (m_g + 1)$ count states (tested through 20 participants).

F. Analytical guarantees

These certificates delimit the logical scope. Under the interior and differentiability assumptions of (14), the threshold in (15) is exact; Experiment 1 checks all 56 cells. The workload set, simplex schedules, and two-sided envelope are feasible. For any execution c ,

$$\begin{aligned} [p_{dt}^{\text{safe}} - c_{dt}]_+ & \leq [p_{dt}^{(\ell^*)} - c_{dt}]_+, \\ [c_{dt} - p_{dt}^{\text{safe}}]_+ & \leq [c_{dt} - p_{dt}^{\text{floor}}]_+ + \varepsilon. \end{aligned}$$

Summation gives the false-credit and under-credit bounds without locked outcomes.

Base-case and stacked N-1 values are convex piecewise-affine functions of nodal demand; (30) gives the linear-value bound. Each dispatch embedded in (38) is feasible, so (39) holds for the five frozen conversion scenarios Ξ_5 . For the continuous audit in Experiment 15, the realized counterfactual is held fixed while the affine segment between the two frozen contract profiles is varied; the joint LP gives the exact segment minimum and convexity puts its maximum at an endpoint. No endpoint assertion is made for a simultaneously conversion-scaled realized counterfactual. Experiments 9 and 15 use independent 40- and four-segment models.

The digest detects post-commit scheduler-tuple changes, while the execution join audits ordering and energy conservation. An optimum of (6) gives declaration-only runtime, capacity, and nameplate feasibility without aggregate rounding. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M , the typed residuals are

$$\begin{aligned} r_j^{\text{job}} &= \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - E_j^{\text{run}}, \\ r_{d\tau}^{\text{agg}} &= X_{d\tau} - \sum_j B_{dj} u_{j\tau}, \\ r_{b\tau}^{\text{map}} &= p_{b\tau} - P_{b\tau}^{\text{fix}} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}. \end{aligned}$$

The first residual uses the same E_j^{run} as (6); $\hat{E}_j$ remains diagnostic. Residuals at most ε identify the witness, checked in Experiments 22 and 27 before N-1 valuation; allocation checks $\sum_i \phi_i = v(N)$ separately. Moreover, if the feasible SCED value is L_V -Lipschitz on the committed polytope, the mapped-value error is bounded by

$$|V(p^{\text{map}}) - V(p^{\text{job}})| \leq L_V \|r^{\text{map}}\|_1. \quad (42)$$

Equation (42) makes the zero-residual certificate explicit: the network solver receives the same nodal profile implied by the indexed jobs, rather than an independently refitted aggregate trace.

IV. EXPERIMENTAL DESIGN

A. Data, preprocessing, and information boundary

BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. The full population contains 75,326 valid submissions and 71,128 positive-energy execution matches, of which 68,662 have valid intervals in the common 121-day tensor used by Experiments 1-13 [2]. Submission time defines arrivals and measured intervals define scoring truth. Chronological workload scaling uses the first 40 complete days. The GPU-power model uses 44,391 training jobs, 9,637 calibration-validation jobs, and 41,154 locked jobs; the calibration-validation ratios define the five scenarios Ξ_5 in (38) without synthetic replacement. Experiment 14 retains all 71,128 matches, whereas Experiment 19 uses every valid declaration with a training-only aggregate entitlement. The submit-time energy-label fit contains 71,141 positive scheduler/DCGM pairs; 71,128 have valid measured intervals for replay, and the 13 label-only pairs cannot enter any decision. Experiment 27 forms the runtime-complete

declaration witness, while execution joins open only for post-event scoring. The runtime window plus a precommitted 96-slot queue allowance defines the contract window, not a submit-to-completion interpretation of Slurm runtime, and membership is never outcome-filtered.

Because the traces lack utility bus locations, four-region coupling is a trace-driven benchmark: a predeclared feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts all 24 permutations, three penetrations, and two concentration controls; Experiment 18 uses fixed generator-bus strata at native-load multiplier 0.90, so its transfer result is conditional on that scenario. Calibration-validation ratios span 0.5827–5.3209 and the locked partition is a generalization audit. Equation (37) carries the fixed 6-MW/site component separately and calibrates payment-network scale on q_{99} at a predeclared 4% peak penetration; both raw endpoints remain activation-eligible under that N–1-feasible scale. The separate flexible-only capacity-safe factor is retained as a planning diagnostic and is not an endpoint clip. The Power Grid Library (PGLib) and MATPOWER/PYPOWER provide topology, ratings, and costs [12], [28]; RTS-24 follows the published case [29].

B. Baselines, parameters, and inference

Experiment 2 compares meter/statistical baselines with complete-ledger quantile/feasible projections, synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier; payment-certified verifier is evaluated separately in Experiments 9 and 15. Feature equality is audited while execution truth is withheld. Deferral windows are 0, 4, and 512 slots; each region has a 118-MW flexible nameplate and 6-MW fixed component. Six metadata projection penalties and a five-point causal Pareto grid are selected by nested contiguous folds; the selected contract uses a 10^{-6} validation RMS non-inferiority certificate based on (26). Normalized total/CVaR preferences and the five-point stress grid are validation-only. The active-window scale in (7) uses submit-time declarations and locked-day overlap; the gate uses slot 62 and commits 3 MWh; payment uses an independent 40-segment evaluator with two date-level workers. The F1 score (harmonic mean of precision and recall) is denoted F_1 throughout the locked scoring tables. Moving three-day bootstrap replicates and all 2^{18} block-sign assignments use Holm correction [30], [31]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s). All release runs clamp BLAS and HiGHS to one numerical-backend thread per worker; the bounded pools use at most 12 workers and remain below the project-wide 20-core ceiling. Every long panel writes a schema- and profile-validated checkpoint before a resumed cell is accepted.

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage/reactive-generation bounds, and apparent-power ratings [12]. After the intact solve, each credible outage fixes every non-reference active output at its intact value, $P_{Gi}^{(k)} = P_{Gi}^{(0)}$, while the reference active balance, reactive outputs, and voltages remain recourse. Apparent-power, voltage, and fixed-plan

deviations are checked after every solve; non-reference bounds are the intact output $\pm 10^{-8}$ MW with a 10^{-6} -MW margin, and the largest deviation is 2.52×10^{-8} MW.

Experiments 1–2 test incentives/workload verification; 3–9 test signed settlement, transfer, risk, and N–1 value; 10–12 test AC/spatial transfer and continuous-cycle remuneration; 13–16 provide measured replay, indexed witness, conversion intervals, and reconciliation. Experiments 17–25 freeze the gate, verify shared-plan AC feasibility, audit scale/coupling, replay an independent event, and enumerate all RTS-24 outages. Experiment 27 adds the runtime-complete common witness; Experiment 28 audits the held-out risk tail with absolute-MWh and normalized-ratio metrics under a fixed block bootstrap; Experiment 29 recomputes the finite risk-to-executable enumeration and its target residual; Experiment 30 audits active-window scaling, information-set lineage, and the closed-meter payment gate. Comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated. Experiment 26 independently recomputes the declaration-to-network residuals and checks the common witness digest, source hashes, and role labels of the risk, payment, and outage artifacts.

V. RESULTS

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (15) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates twelve methods on the locked 54-day panel (Table I). The causal Pareto risk contract reaches $\text{nRMSE}/F_1/\text{false credit}/\text{under-credit}$ of 0.691/0.699/1.895/17.024 MWh/day. It strictly improves all four locked metrics over the single feasible projection (0.943/0.463/2.470/26.812) and weakly dominates every other declaration-causal, workload-feasible comparator in the table. The selected profile is the exact one-hot causal Pareto vertex returned by the joint convex program; this is expected because the 10^{-6} RMS certificate makes that validation-selected vertex the unique admissible minimum-RMS solution. The validation certificate records reference and fitted MSE 188.931512 versus an upper bound of 188.931701, an RMS bound of 13.745236 against 13.745243, zero KKT and primal residuals, and satisfied total and daily-CVaR budgets. The ex-post rows use complete execution information and are retained as information-boundary diagnostics, while High-5-of-10 is not workload-feasible; neither is an admissible declaration-time comparator. Total-only, CVaR-only, and joint ablations return the same certified vertex, so the two contractual axes are verified without introducing a post-solution heuristic. The independent meter replay remains separate: it supplies observational scoring and never enters fitting or reserve selection.

Three-day moving-block intervals and paired tests are reported for each risk axis in Experiment 28. Their comparison

TABLE I

LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS). THE PROPOSED CONTRACT IS BOLD; EX-POST ROWS ARE INFORMATION-BOUNDARY DIAGNOSTICS AND HIGH-5-OF-10 IS NOT WORKLOAD-FEASIBLE.

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	1.586	41.220	11.169	0.572
Ridge	1.038	21.906	25.226	0.545
Gradient Boosting	1.011	20.977	26.401	0.534
Extra Trees	1.090	26.629	18.176	0.597
Metadata Gradient Boosting	0.774	9.183	21.528	0.640
Ex-post Metadata Gradient Boosting	0.750	9.171	20.817	0.643
Ex-post Quantile Gradient Boosting	1.337	0.591	54.391	0.237
Synthetic Control	0.782	9.394	21.678	0.676
Feasible Quantile Projection	1.344	0.098	52.647	0.225
Single Feasible Projection	0.943	2.470	26.812	0.463
Tail-Risk Feasible Counterfactual	0.691	1.895	17.024	0.699
Risk-Constrained Convex Verifier	0.691	1.895	17.024	0.699

TABLE II

BASE-CASE AND COMPLETE N-1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	—	—
Base case	Nodal exact net value	151.719	455.865	—	—
N-1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N-1	N-1 linear	2.208	150.461	0.60/1.91	113.43/20.38
N-1	N-1 exact	0.990	150.955	0.87/0.01	113.43/20.18

set is declaration-causal and workload-feasible; complete-ledger ex-post baselines remain outside that set by construction. This separation makes the information boundary explicit while preserving the observed four-metric dominance claim within the deployable set.

The independent trace-meter replay is a deployment check: event-window MAE is 1.909/5.302/4.813 MW for the feasible-quantile comparator/tail-risk counterfactual/risk-constrained veriferover the same 54 days (Experiment 20), separate from mechanism-isolation scoring. Closest domain translations use identical arrivals, deadlines, capacities, event slots, and days: the rolling-horizon ledger gives nRMSE/false-credit/ F_1 0.943/2.470/0.463, the batch-flexibility translation gives 1.357/0/0.183, and the coupled-service translation gives 1.376/0/0.111. These controls translate the published structures into the same exact ledger and are labelled translations rather than software reproductions [5], [10]. The ablation panel contains 810 globally solved schedules; its binding-share and marginal-cost diagnostics are reported in the archived certificate.

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms; Table II reports the trace-anchored locked rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response, with workload errors \$455.865 and \$421.677. Signed netting reduces trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$); uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$), and linear-to-exact value by \$0.671 ($p = 0.8026$). All 5,616 interval/counterfactual instances satisfy (30), with maximum residual 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N-1 security evidence

Experiments 5–8 establish the cross-network and N-1 replay. Across 6,480 four-network outcomes, exact value improves trace error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05) with a \$0.050/day workload change. The capacity/deadline and coalition panels solve 810 schedules and 1,568 exact participant states, respectively, with residuals below 5.68×10^{-14} dollars. The 37-outage RTS-24 replay enforces (36); trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for the three settlements, with 1.000 p.u. maximum loading.

Table III summarizes the complete replay scope.

Experiment 9 tests (39) on 54 programs with a validation-frozen seven-profile hull. The cap reference ($\rho = 10$) and payment target ($\rho = 3$) are selected by contiguous folds; the five declared scaling factors are 0.583/0.687/0.970/1.349/5.321 and all certificate violations are below 3×10^{-11} USD. PaymentSafe/single/feasible-quantile/RiskSafe q50/q99 MAE is 56.858/321.660, 40.675/231.475, 55.360/312.933, and 31.653/178.847 USD/day, respectively. On the two held-out interior transfer scenarios, RiskSafe has 10.440/15.688 USD/day MAE versus 9.981/14.999 for PaymentSafe; these rows and the moving-block intervals remain outside target selection.

The AC replay (Experiment 10) has 864 connected-base and 5,400 corrective IEEE-9/14 cells, all feasible, while Experiment 18 adds 9,648 admissible contingencies and 144 intact solves with zero voltage violation. Experiment 11 evaluates 26 assignments at three penetrations; its MAE triples are \$101.155/\$203.361/\$303.195, \$105.250/\$211.555/\$315.490, and \$81.613/\$164.249/\$244.481 for RiskSafe, single, and feasible quantile. Experiment 16 reports calibration $R^2 = 0.8956/0.8196$; Experiment 12 reports event-window MAE \$26.937/\$34.166/\$8.659/\$5.325 and complete-cycle residual \$0.621/\$0.569/\$0.104/\$0.

D. Independent trace and deployment-boundary audits

The immutable scheduler/DCGM replay gives a 71,128-job residual below 10^{-17} MWh and no event tag.

Experiment 17 gives the complete-ledger comparator (0.691/1.895/0.699) and slot-62 gate (1.366/0.494/0.080); the gate is fixed pre-scoring, masks post-gate arrivals and execution truth, and requires 3 MWh on 54 days. The comparator is therefore not a closed-meter payment. Experiment 20 gives 4.813 MW trace-meter MAE, and Experiment 23 gives 0.0210/0.0194/ $F_1 = 0.529$ under a separate tariff, with exact $F_1 = 1.000$ on independent replay. Table III records the boundary.

E. Indexed workload counterfactual

Experiment 27 assigns all 75,326 valid submissions a runtime-complete block; Exp29 enumerates 7,306,622 feasible starts and reproduces every minimizer (Table III). Name-plate/central entitlements are 1,972.115/0.570 MWh, active declaration energy on the locked windows is 655.753 MWh, and all energy-conservation residuals are below 10^{-11} MWh.

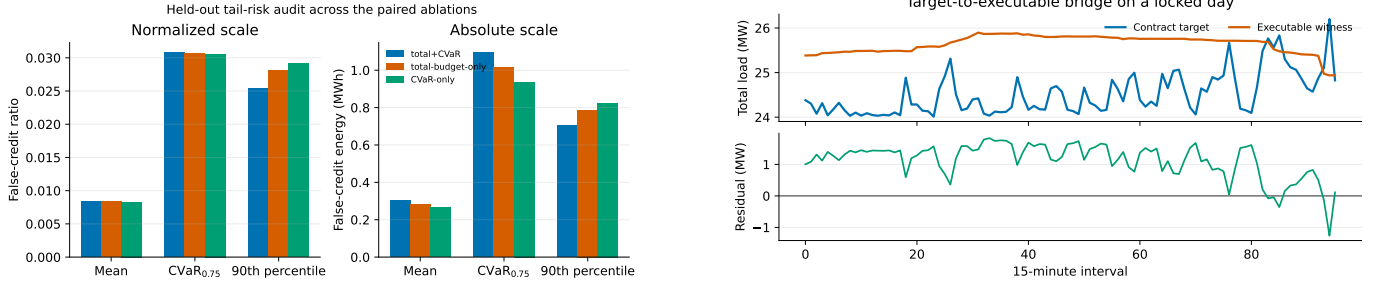


Fig. 3. Result evidence for the two coupled certificates. Left: held-out normalized and absolute false-credit tails for the joint, total-only, and CVaR-only fits. Right: the locked-day aggregate contract target and the declaration-indexed executable realization, with its audited residual below; the realization is the exact finite-start policy minimizer, while target tracking is audited rather than assumed.

TABLE III

DECLARED INFORMATION-BOUNDARY, SECURITY-REPLAY, AND INDEXED-CERTIFICATE COVERAGE. TRIPLETS ARE NRMSE/FALSE RESPONSE/ F_1 ; DECLARATION CHECKS PRECEDE LOCKED SCORING.

Panel	Cells and certificate
Complete-ledger risk contract	54 days; 0.691/1.895/0.699
Slot-62 gate-causal contract	54 days; 1.366/0.494/0.080; no future arrivals
Contract-lineage audit	9/9 checks; no future arrivals; payment closed-meter gate
Indexed job witness	75,326/71,128 jobs; 13,198,247 variables; 7,306,622 starts; residual < 10^{-11} MWh
Executable bridge/network replay	nRMSE 0.03631/relative-L2 1.410; signed/cash 174.666/264.481 USD; 947 debit cells
Frozen-profile DC N-1 (RTS-24)	864 cells; 37 outages/cell; 1.0000 p.u.
Shared-plan AC (IEEE-9)	3,888; 0.880/0.907/0.935 p.u.
Corrective AC (IEEE-9/14)	864 base + 5,400 outages; all feasible
Admissible fixed-plan AC N-1 (four net-works)	9,648 outages + 144 intact; certified

Event reduction is 4.68225 MWh. The submit-time-scaled risk-constrained verifier bridge has total nRMSE 0.03631 and flexible relative-L2 residual 1.410 under (8); its maximum absolute residual is 4.449 MW. The 5,184-cell full-day ledger retains 947 signed debit cells and reports a signed network replay value of -0.686 USD, a signed cycle value of 174.666 USD, and a cash-floor presentation of 264.481 USD. Exp30 marks this replay as declaration-only: the independent observed-meter panel supports scoring, while a closed meter is required before payment activation.

VI. CONCLUSION

This paper couples a provenance-bound workload verifier with signed nodal valuation. The locked causal Pareto contract reaches nRMSE/false credit/ $F_1=0.691/1.895/0.699$ while satisfying its predeclared total and daily-tail budgets and weakly dominating the declaration-causal workload-feasible comparison set (with strict gains over the single feasible projection; Fig. 3). The witness covers 75,326 submissions, 5,184 RTS-24 cells, and 655.753 MWh of active declaration energy. The exact start audit reports total nRMSE 0.03631 and flexible relative-L2 residual 1.410. The signed network replay retains negative cells and is reported separately from the independent meter score; it is a feasibility/value certificate rather than a closed-meter payment. Deployment requires scheduler, telemetry, and topology attestation, together with a payment gate that opens only after the closed meter is available.

REFERENCES

- [1] Y. Wang, Y. Chen, Z. Li, X. Kang, Y. Fang, Y. Zhou, Y. Zheng, Z. Tang, X. He, R. Guo, X. Wang, Q. Wang, A. C. Zhou, and X. Chu, "BurstGPT: A real-world workload dataset to optimize large language model serving systems," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5831–5841, 2025.
- [2] S. Samsi, M. L. Weiss, D. Bestor, B. Li, M. Jones, A. Reuther, et al., "The mit supercloud dataset," in *2021 IEEE High Performance Extreme Computing Conference*, pp. 1–8, 2021.
- [3] California Independent System Operator, "California iso baseline accuracy work group final proposal," tech. rep., California ISO, 2017.
- [4] X. Wang and W. Tang, "Modeling and analysis of baseline manipulation in demand response programs," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1178–1186, 2022.
- [5] Y. Cao, M. Cheng, S. Zhang, H. Mao, P. Wang, C. Li, Y. Feng, and Z. Ding, "Data-driven flexibility assessment for internet data center towards periodic batch workloads," *Applied Energy*, vol. 324, p. 119665, 2022.
- [6] Y. Cao, F. Cao, Y. Wang, J. Wang, L. Wu, and Z. Ding, "Managing data center cluster as non-wire alternative: A case in balancing market," *Applied Energy*, vol. 360, p. 122769, 2024.
- [7] W. Zhang, L. A. Roald, A. A. Chien, J. R. Birge, and V. M. Zavala, "Flexibility from networks of data centers: A market clearing formulation with virtual links," *Electric Power Systems Research*, vol. 189, p. 106723, 2020.
- [8] W. Zhang and V. M. Zavala, "Remunerating space-time, load-shifting flexibility from data centers in electricity markets," *Applied Energy*, vol. 326, p. 119930, 2022.
- [9] W. Zhang, L. A. Roald, and V. M. Zavala, "Exploring the impacts of power grid signals on data center operations using a receding-horizon scheduling model," *IEEE Transactions on Sustainable Computing*, vol. 8, no. 2, pp. 245–256, 2023.
- [10] M. Chen, C. Gao, M. Shahidehpour, Z. Li, S. Chen, and D. Li, "Internet data center load modeling for demand response considering the coupling of multiple regulation methods," *IEEE Transactions on Smart Grid*, vol. 12, no. 3, pp. 2060–2076, 2021.
- [11] B. Stott, J. Jardim, and O. Alsac, "DC power flow revisited," *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009.
- [12] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [13] D. A. Tejada-Arango, P. Sánchez-Martin, and A. Ramos, "Security constrained unit commitment using line outage distribution factors," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 329–337, 2018.
- [14] C. Li, K. Zheng, H. Guo, C. Kang, and Q. Chen, "Computation-power coupled modeling for idcs and collaborative optimization in adms," *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 2762–2775, 2024.
- [15] National Institute of Standards and Technology and Q. Dang, "Secure hash standard," Tech. Rep. FIPS PUB 180-4, National Institute of Standards and Technology, 2015.
- [16] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
- [17] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [18] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [19] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006.
- [20] A. Abadie, A. Diamond, and J. Hainmueller, "Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program," *Journal of the American Statistical Association*, vol. 105, no. 490, pp. 493–505, 2010.
- [21] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [22] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.
- [23] B. Satchidanandan, M. Roozbehani, and M. A. Dahleh, "A two-stage mechanism for demand response markets," *IEEE Control Systems Letters*, vol. 7, pp. 49–54, 2023.
- [24] M. Chen, C. Gao, M. Shahidehpour, and Z. Li, "Incentive-compatible demand response for spatially coupled internet data centers in electricity markets," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3056–3069, 2021.
- [25] J. John F. Nash, "The bargaining problem," *Econometrica*, vol. 18, no. 2, pp. 155–162, 1950.
- [26] North American Electric Reliability Corporation, "TPL-001-5.1: Transmission system planning performance requirements," tech. rep., North American Electric Reliability Corporation, 2020.
- [27] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games, Volume II* (H. W. Kuhn and A. W. Tucker, eds.), pp. 307–317, Princeton University Press, 1953.
- [28] S. Babaeinejadarsookolae, A. Birchfield, R. D. Christie, C. Coffrin, C. DeMarco, R. Diaio, et al., "The power grid library for benchmarking ac optimal power flow algorithms," *arXiv preprint arXiv:1908.02788*, 2019.
- [29] C. Grigg, P. Wong, P. Albrecht, R. Allan, M. Bhavaraju, R. Billinton, Q. Chen, C. Fong, S. Haddad, S. Kuruganty, et al., "The IEEE reliability test system—1996," *IEEE Transactions on Power Systems*, vol. 14, no. 3, pp. 1010–1020, 1999.
- [30] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.
- [31] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.